



CIVITAS indicators

Share of logistics operators participating in collaborative schemes for urban freight delivery (TRA_FR_CL)

DOMAIN

 Transport	 Environment	 Energy	 Society	 Economy
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TOPIC

Freight

IMPACT

Collaborative logistics

Increasing the number of operators participating in collaborative schemes

TRA_FR

Category

Key indicator	Supplementary indicator	State indicator
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CONTEXT AND RELEVANCE

Urban freight transport generates significant externalities in cities, including congestion, air pollution, greenhouse gas emissions, noise, road safety risks, and disproportionate occupation of scarce public space. These impacts are exacerbated by structural inefficiencies in distribution systems, such as sub-optimal routing, low vehicle load factors, and multiple operators serving overlapping delivery areas. Promoting collaboration and coordinated optimisation among logistics providers through measures such as shared consolidation facilities, joint route planning, and data sharing can address these inefficiencies by consolidating shipments, increasing load factors, and reducing redundant trips.

This indicator measures the number of logistics operators in the pilot city participating in collaborative optimisation schemes. **It is a relevant indicator when the policy action aims to increase the efficiency of urban freight distribution through the promotion of collaborative optimisation schemes. A successful action is reflected in a HIGHER value of the indicator.**

DESCRIPTION

This indicator represents the number of logistics operators in the experiment city participating in collaborative optimisation schemes for urban freight delivery. These aim to facilitate cooperation between logistics providers, by setting up networks that enable different operators to share data and consolidate shipments, increasing load efficiency and reducing distances travelled in urban area.

The unit of measurement is **percentage (%)**.

METHOD OF CALCULATION AND INPUTS

The indicator should be computed **by the user** by applying the method described and then coded in the supporting tool.

Method	
Estimation of the index based on direct observation	Significance: 0.50 
INPUTS The following information is needed to compute the indicator: a) Total number of logistics operators in the pilot city. b) Number of logistics operators participating in collaborative optimisation schemes. The experiment is expected to increase the number of operators participating in collaborative schemes.	
METHOD OF CALCULATION The indicator is calculated by the user as the ratio of participating operators to the total number of operators.	

EQUATION

The indicator is calculated using the following equation:

$$ShColOp = \frac{ColOp}{Op} \times 100$$

Where:

ColOp = Logistics operator participating in collaborative schemes

Op = Total logistics operators in the pilot city

ALTERNATIVE INDICATORS

This indicator measures the share of operators in the pilot city participating in collaborative optimisation schemes. To assess the efficiency of freight transport in terms of the average load factor of delivery vehicles, the framework provides indicators **TRA_FR_EF4** (B2B deliveries) and **TRA_FR_EF5** (B2C deliveries).